MSE 311 Thermodynamics of materials-Quiz 9

Name: *Student ID:*

1. (a) Write down the mixture Gibbs free energy (ΔG^M) in a binary system (A and B) under regular solution. (b) Then write down its equation of binodal curve ($\partial \Delta G^M / \partial X_A$) and equation of spinodal curve ($\partial^2 \Delta G^M / \partial X_A^2$).

$$\Delta \boldsymbol{G}^{\boldsymbol{M}} = \Delta \boldsymbol{G}^{\boldsymbol{M},id} + \boldsymbol{G}^{\boldsymbol{XS}} = \boldsymbol{RT}(\boldsymbol{X}_{\boldsymbol{A}} \ln \boldsymbol{X}_{\boldsymbol{A}} + \boldsymbol{X}_{\boldsymbol{B}} \ln \boldsymbol{X}_{\boldsymbol{B}}) + \Omega \boldsymbol{X}_{\boldsymbol{A}} \boldsymbol{X}_{\boldsymbol{B}}$$

$$\frac{\partial \Delta \boldsymbol{G}^{\boldsymbol{M}}}{\partial \boldsymbol{X}_{\boldsymbol{A}}} = \Omega(1-2\boldsymbol{X}_{\boldsymbol{A}}) + \boldsymbol{RT} \ln\left(\frac{\boldsymbol{X}_{\boldsymbol{A}}}{\boldsymbol{X}_{\boldsymbol{B}}}\right)$$

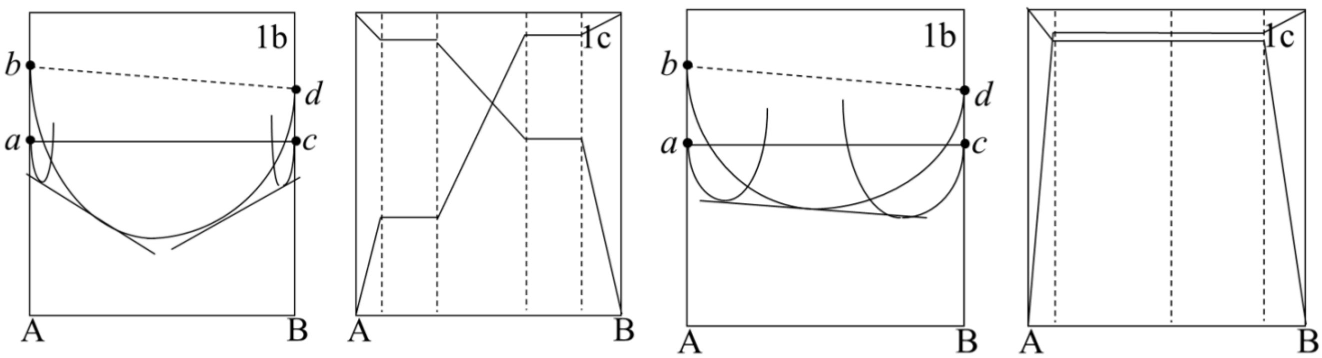
$$\frac{\partial^2 \Delta \boldsymbol{G}^{\boldsymbol{M}}}{\partial \boldsymbol{X}_{\boldsymbol{A}}^2} = -2\Omega + \boldsymbol{RT}\left(\frac{1}{\boldsymbol{X}_{\boldsymbol{A}}} + \frac{1}{\boldsymbol{X}_{\boldsymbol{B}}}\right) = 0$$

2. Write down the equation of the critical temperature in a binary system under the regular solution and explain its physical meaning.

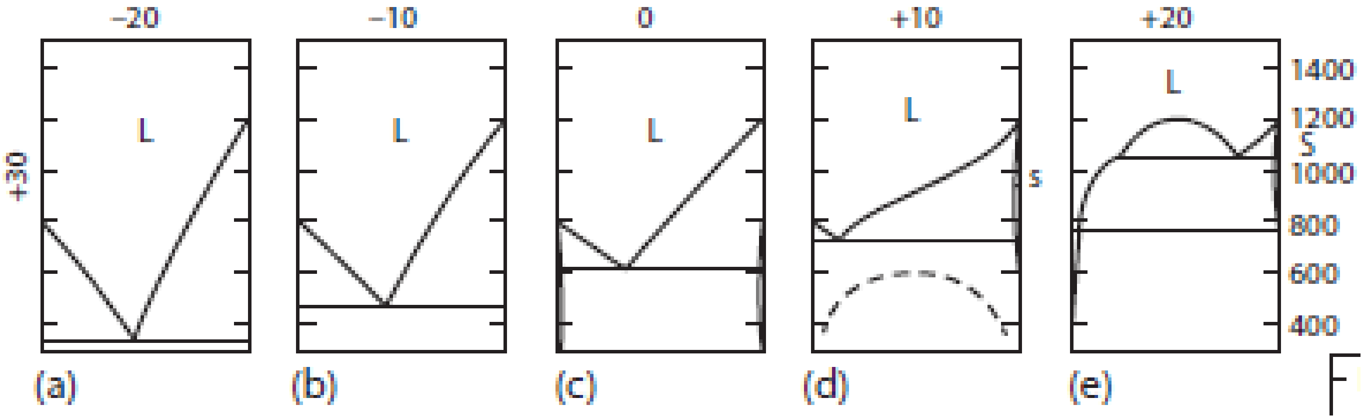
$$T_{cr} = \frac{\Omega}{2R}$$

It's the critical temperature between the single phase and the phase separation.

3. Draw the activities of the components of the system A – B according the molar Gibbs free energies of mixing in the right figure.



4. Please indicate the trend of Ω_l in the following five figures and give your reasons.



From the left to right, the liquid solutions become increasing less stable relative to the solid phases with the consequence that the eutectic temperature is increased. Thus, the Ω_l becomes more larger.